



A Comprehensive Data Analytics Study of Used Cars Prices in India

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CHAPTER 1

INTRODUCTION

1.1 Project Objective

Buying a used car is a significant decision that many people in India make each year. With numerous car brands, models, age groups, and price ranges available in the market, it can be challenging for buyers to make informed decisions. The objective of this project is to analyze the variation in used car prices based on different car brands and to identify the best quality cars available at reasonable prices. By utilizing data analytics techniques, this project aims to provide valuable insights to potential used car buyers in India.

1.2 Overview of Project

This project uses web scraping, data cleaning, and visualization techniques to collect and analyze information about used cars available in the Indian market. The study covers various aspects including price variations across different brands, the relationship between car age and price, the impact of kilometers driven on pricing, and differences in prices based on fuel type and transmission. Interactive visualizations are used to present the findings, helping users explore the data and draw meaningful conclusions.

This project adheres to a methodical approach that comprises:

- ❖ Data collection: taking information from the Spinny website, a reputed used car selling platform in India
- ❖ Preparing the dataset for analysis involves cleaning and organising it.
- ❖ Identifying patterns and trends in car prices based on various factors
- ❖ Dashboards and Visualisation: Developing interactive Power BI dashboards to effectively communicate results.

1.3 Problem Statement

Choosing the right used car in India is challenging due to the lack of structured and comparative information. Several factors influence the decision, including price, brand reputation, car age, mileage, fuel type, and transmission type. Buyers often rely on limited information from multiple sources, making the process overwhelming and potentially leading to poor decisions.

The absence of standardized data makes it difficult to compare cars based on objective criteria. Price is a crucial factor, as it varies significantly between different brands, models, and conditions. Additionally, the relationship between a car's age, kilometers driven, and its price is not always clear to buyers.

This project takes a data-driven approach, collecting and analyzing used car data from reliable sources. By leveraging various visualization techniques, it simplifies the comparison process and helps potential buyers make informed decisions.

1.4 Scope

The study covers various factors related to used cars in the Indian market, such as:

- **Price Analysis:** Examining how prices vary across different car brands, models, and cities
- **Age Impact:** Analyzing the relationship between a car's age and its market value
- **Kilometers Driven:** Understanding how the distance driven affects a car's price
- **Fuel Type Comparison:** Comparing prices between petrol, diesel, and other fuel types
- **Transmission Type:** Analyzing price differences between manual and automatic transmissions
- **EMI Analysis:** Exploring monthly payment options for different price ranges

The findings from this project will be useful for multiple stakeholders:

- **Car Buyers:** To make better decisions when purchasing used cars based on value for money
- **Car Sellers:** To understand optimal pricing strategies for different types of cars
- **Market Analysts:** To identify trends and patterns in the used car market
- **Financial Institutions:** To better understand the market when offering auto loans

CHAPTER 2

DATA COLLECTION AND SOURCES

Data source

The primary data source for this project is Spinny, one of India's leading online platforms for buying and selling used cars. This website provides comprehensive information about used cars listed for sale, including details such as car model, manufacturing year, price, kilometers driven, fuel type, and other specifications.

1. Extracting College Details

To ensure a comprehensive dataset, the following key attributes were extracted:

- Car Name (Brand and Model)
- City and State (Location)
- Manufacturing Year
- Price
- Kilometers Driven
- Transmission Type (Manual or Automatic)
- Fuel Type (Petrol, Diesel, CNG, Electric, etc.)
- EMI per Month

These details were extracted using BeautifulSoup in Python, where specific HTML elements containing car information were parsed. Additional attributes like the car's age were calculated based on the manufacturing year.

2. Web Scraping Process

To efficiently extract the data, the following tools and libraries were utilized:

- BeautifulSoup & Requests: For retrieving and parsing webpage content
- Pandas & NumPy: For data structuring and preprocessing
- Regular Expressions (Regex): For extracting specific patterns from text

Steps in the Web Scraping Process:

1. **Sending Requests:** HTTP requests were sent to Spinny to access used car listing pages across different cities in India.
2. **Parsing HTML Content:** BeautifulSoup was used to identify and extract relevant HTML elements containing car data.
3. **Extracting Data:** The parsed data was structured into DataFrames using Pandas, including cleaning and preprocessing steps.
4. **Saving Data:** The final dataset was saved in CSV format for further analysis and visualization.

3. Challenges in Data Collection

Several challenges were encountered during the data collection process:

- **Dynamic Web Content:** CarWale uses JavaScript to load content dynamically, requiring the implementation of advanced scraping techniques.
- **Rate Limiting:** The website has mechanisms to prevent excessive scraping, necessitating the use of delays between requests.
- **Inconsistent Data Formats:** Information like prices and kilometers driven appeared in different formats across listings.
- **Missing Information:** Not all car listings had complete information, resulting in some missing values.
- **Large Data Volume:** Managing the large volume of data collected from thousands of car listings required efficient storage and processing systems.

CHAPTER 3

DATA PREPROCESSING

The raw data collected through web scraping required extensive cleaning and preprocessing before it could be analyzed effectively. This preprocessing was done using Python libraries like Pandas and NumPy, with the following steps:

3.1 Cleaning and Formatting

Several preprocessing steps were applied to clean and format the collected data:

Steps Taken for Data Cleaning:

1. Duplicate Removal:

- Identified and removed duplicate car listings to ensure uniqueness in the dataset.

2. Handling Missing Values:

- Analyzed missing values in key columns such as price, kilometers driven, and fuel type.
- Applied appropriate strategies like removal or imputation based on the nature of the missing data.

3. Data Type Conversion:

- Converted price and kilometers driven from string format (with commas and units) to numeric format.
- Created a new 'Age' column by calculating the difference between the current year and the manufacturing year.
- Standardized text fields like car names and locations for consistent analysis.

4. Outlier Detection and Treatment:

- Identified outliers in price and kilometers driven using statistical methods.
- Applied appropriate treatments to outliers to prevent them from skewing the analysis.

3.2 Feature Engineering

Additional features were created to enhance the analysis:

1. Age Category:

- Categorized cars into age groups (0-3 years, 3-5 years, 5-10 years, 10+ years).
- 2. **Price Range:**
 - Created price range categories to facilitate segment-based analysis.
- 3. **Brand Extraction:**
 - Extracted the car brand from the full car name to enable brand-based analysis.
- 4. **KM Driven Ranges:**
 - Created kilometer ranges to analyze price variation based on usage levels.

3.3 Data Integration

After preprocessing, the cleaned dataset was ready for analysis and visualization:

- The final dataset contained 10,241 unique car listings with 11 attributes.
- All numeric columns were properly formatted and ready for statistical analysis.
- Text fields were standardized for consistent grouping and filtering.

CHAPTER 4

DATA ANALYSIS AND VISUALIZATION

4.1 Overview of Analysis

The analysis of the preprocessed data focused on five key areas:

1. **Brand-Based Price Analysis:** Examining how prices vary across different car brands
2. **Age vs. Price Relationship:** Analyzing the correlation between a car's age and its price
3. **Kilometers Driven Analysis:** Understanding the impact of usage on car prices
4. **Fuel Type Comparison:** Comparing prices and availability across different fuel types
5. **Geographic Distribution:** Analyzing price variations across different cities and states

Each analysis area employed appropriate statistical methods and visualization techniques to extract meaningful insights from the data.

4.2 Key Visualizations and Insights

4.2.1 Brand-Based Price Analysis

Visualizations:

- Bar chart comparing average prices across top car brands
- Box plots showing price distributions for each brand
- Scatter plot of price vs. brand with color coding for car age

Insights:

- Luxury brands like Mercedes-Benz, BMW, and Audi have the highest average prices
- Maruti Suzuki, Hyundai, and Tata offer the most affordable used car options
- Even within the same brand, prices vary significantly based on model and specifications
- Some mid-range brands like Honda and Toyota maintain better value over time compared to others

4.2.2 Age vs. Price Relationship

Visualizations:

- Line chart showing the average price decline over car age
- Scatter plot of age vs. price with trend line
- Heatmap showing the relationship between brand, age, and price

Insights:

- Cars lose approximately 15-20% of their value in the first year
- The depreciation rate slows down after 5 years of age
- Luxury brands tend to depreciate faster in absolute terms but slower in percentage terms
- Some brands maintain their value better than others as they age

4.2.3 Kilometers Driven Analysis

Visualizations:

- Scatter plot of kilometers driven vs. price
- Box plots showing price distribution across kilometer ranges
- Line chart showing the average price decline with increasing kilometers

Insights:

- Cars with lower kilometers driven command premium prices
- The price drop is steeper for the first 50,000 kilometers
- After 100,000 kilometers, the impact of additional kilometers on price diminishes
- The effect of kilometers driven varies by brand and fuel type

4.2.4 Fuel Type Comparison

Visualizations:

- Bar chart comparing average prices across fuel types

- Pie chart showing the distribution of fuel types in the market
- Side-by-side bar chart comparing fuel type distribution across different price ranges

Insights:

- Diesel cars are generally priced higher than petrol cars of the same age and brand
- CNG vehicles offer the best value for money in terms of fuel efficiency
- Electric and hybrid vehicles are still rare in the used car market
- Fuel type preference varies significantly across different cities

4.2.5 Geographic Distribution Analysis

Visualizations:

- Map of India showing average car prices by state
- Bar chart comparing average prices across major cities
- Heatmap showing the relationship between city, car brand, and price

Insights:

- Metropolitan cities like Mumbai, Delhi, and Bangalore have higher average prices
- Certain brands are more popular in specific regions
- Tier-2 cities often offer better value for the same car models
- The variety of available cars differs significantly by location

4.3 EMI Analysis

Visualizations:

- Scatter plot showing the relationship between car price and EMI
- Bar chart comparing average EMIs across different brands
- Line chart showing EMI variations across different loan tenures

Insights:

- EMIs are directly proportional to car prices
- EMI options make luxury cars more accessible through monthly payments
- Different financing options have significant impact on total ownership cost
- EMIs for same-priced cars can vary based on interest rates and down payments

4.4 Interactive Dashboard Highlights

An interactive dashboard was developed to allow users to explore the findings dynamically. Key features of the dashboard include:

1. **Car Explorer:** Interactive interface allowing users to filter cars by brand, price range, age, and fuel type
2. **Price Comparison Tool:** Side-by-side comparison of prices across different brands and models
3. **Value Analyzer:** Dynamic visualizations showing value retention across different car types
4. **EMI Calculator:** Tool to estimate monthly payments based on car price and loan terms
5. **Regional Insights Panel:** Summarized statistics for selected cities or states

The dashboard incorporates filters and drill-down capabilities to enable users to navigate from high-level overviews to detailed information about specific car segments.

CHAPTER 5

TOOLS AND TECHNOLOGIES

5.1 Software Used

To ensure efficient data collection, preprocessing, analysis, and visualization, multiple software tools were utilized throughout the project. These tools enabled the seamless handling of large datasets, effective data transformation, and interactive visualization of key insights.

- **Python :** The primary programming language used for data collection, preprocessing, and analysis. Jupyter Notebook provided an interactive environment for executing Python scripts, visualizing data, and debugging efficiently.
- **Power BI:** Used for data visualization and dashboard creation, allowing for the graphical representation of trends in college distribution, ratings, fees, and course offerings. Power BI's interactive dashboards made the insights more accessible to students, educators, and policymakers.
- **Excel:** Played a crucial role in manual data validation, formatting, and cross-referencing information. Excel was particularly useful for quick checks on scraped data, handling small-scale corrections, and verifying outputs before final analysis.

5.2 Python Libraries Used

Several Python libraries were used to automate tasks and manage the data pipeline efficiently:

- **BeautifulSoup & Requests:** Used for extracting college data from the web.
- **Pandas & NumPy:** Assisted in handling large datasets, performing calculations, and manipulating tabular data.
- **Regex:** Employed for pattern matching and extracting specific information from text.

5.3 Power BI Features Utilized

Power BI was extensively used to create interactive dashboards with the following features:

- **Data Modeling:** Creating relationships between datasets to generate meaningful insights.

- **DAX (Data Analysis Expressions):** Used for creating calculated fields and aggregating measures.
- **Drill-Through and Filter Options:** Allowed users to explore detailed insights about engineering colleges.
- **Custom Visuals:** Implemented bar charts, pie charts, line graphs, and maps to display trends effectively.

5.4 Integration of Tools

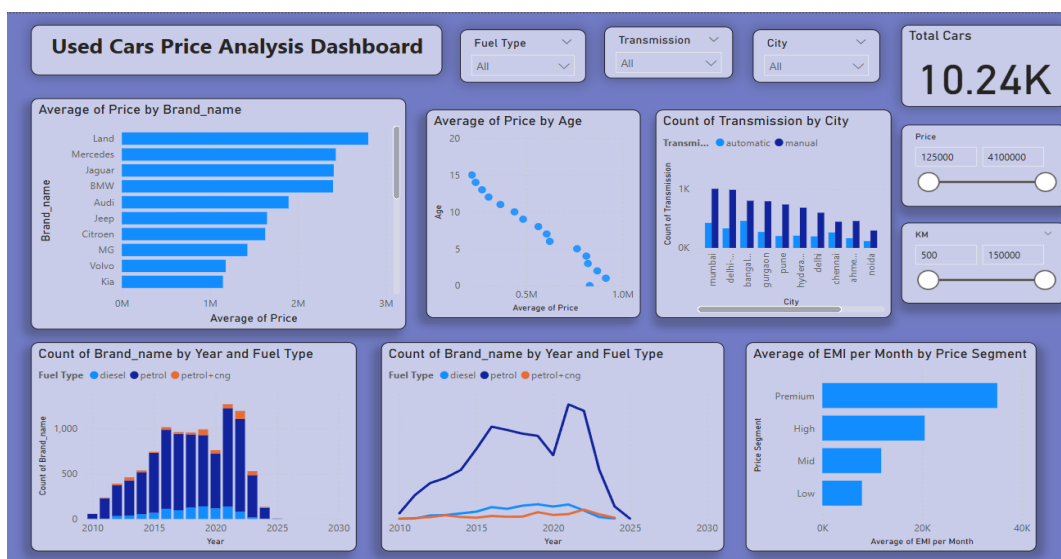
The overall workflow of the project was structured as follows:

1. **Data Collection:** Web scraping using Python libraries.
2. **Data Cleaning:** Performed using Pandas, NumPy.
3. **Data Storage:** Cleaned data stored in CSV/Excel format.
4. **Data Visualization:** Implemented in Power BI with interactive dashboards.

CHAPTER 6

DASH BOARDS

The used car market represents a significant economic sector with distinctive dynamics driven by consumer preferences, economic conditions, and vehicle characteristics. Our comprehensive analysis dashboard examines over 10,000 vehicles across multiple dimensions, including pricing trends across various brands, the correlation between vehicle age and value, geographic distribution patterns, and the impact of features such as transmission type and fuel variants on market positioning. By synthesizing this information through intuitive data visualization tools, we offer stakeholders—dealers, buyers, investors, and market analysts—valuable insights into market segmentation, price determinants, and consumer behavior in the used car ecosystem, highlighting premium segment performance, regional variations in inventory distribution, and the shifting landscape of fuel preferences over the past decade.



Dashboard 1 (Main Overview)

The main dashboard displays a purple/blue themed interface with multiple interactive visualizations:

1. Header Information:

- Title: "Used Cars Price Analysis Dashboard"
- Total Cars: 10.24K vehicles in the dataset

- Filter options for Fuel Type, Transmission, and City

2. **Brand Price Analysis:**

- Bar chart showing average price by brand
- Land Rover appears as the most expensive brand, followed by Mercedes, Jaguar, and BMW
- Lower-priced brands include Volvo and Kia

3. **Price vs. Age Correlation:**

- Scatter plot showing the inverse relationship between car age and price
- Clear downward trend indicating older cars have lower average prices

4. **Transmission Distribution by City:**

- Bar chart showing automatic vs. manual transmission counts across different cities
- Mumbai appears to have the highest number of vehicles
- Most cities show more automatic transmissions than manual

5. **Brand Distribution Over Time:**

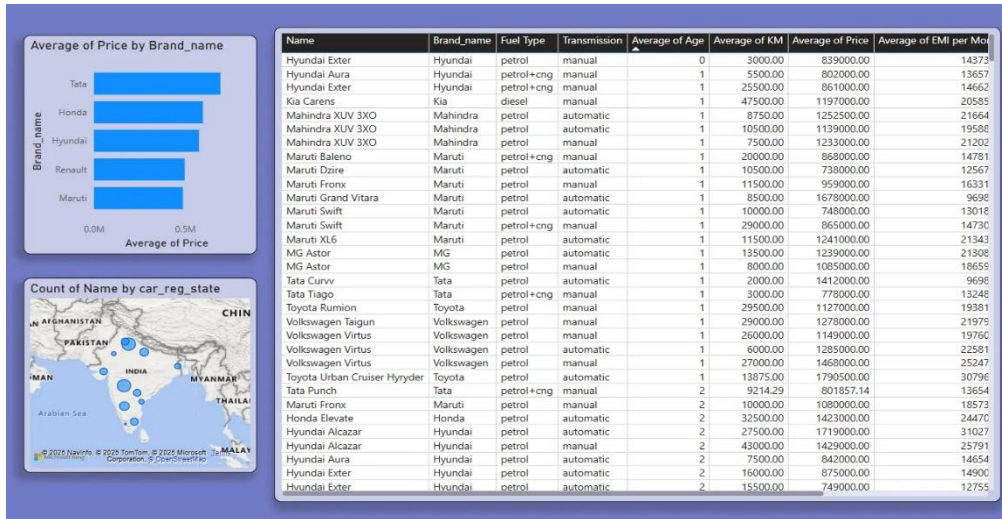
- Two complementary visualizations showing count of brands by year and fuel type
- Left chart: Stacked bar chart showing peak car counts around 2018-2020
- Right chart: Line graph showing the same data with diesel, petrol, and petrol+CNG categories
- Petrol vehicles (blue line) dominate the market

6. **EMI Analysis:**

- Bar chart showing average EMI per month by price segment
- Premium segment has significantly higher EMIs, followed by High, Mid, and Low segments

7. **Interactive Filters:**

- Price range slider (125,000 to 4,100,000)
- Kilometer range slider (500 to 150,000)



Dashboard 2 (Detailed View)

The second dashboard provides a more detailed and granular view:

1. Brand Price Comparison:

- Bar chart showing average price by brand in the Indian market
- Brands include Tata, Honda, Hyundai, Renault, and Maruti

2. Geographic Distribution:

- Map of India showing car registration distribution by state
- Blue circles of varying sizes indicate concentration of registered vehicles
- Highest concentration appears in major urban centers

3. Detailed Data Table:

- Comprehensive table showing individual car models with detailed specifications
- Columns include: Name, Brand name, Fuel Type, Transmission, Average Age, Average KM, Average Price, Average EMI per Month
- Models from various manufacturers including Hyundai, Kia, Mahindra, Maruti, MG, Tata, Toyota, and Volkswagen
- Data shows different fuel types (petrol, diesel, petrol+CNG) and transmission types (manual, automatic)
- Age data shows mostly newer cars (0-2 years old)
- Wide price range from approximately 738,000 to 1,795,000 currency units

CHAPTER 7

CONCLUSION

7.1 Key Findings

This comprehensive analysis of the used car market in India has revealed several significant insights:

- **Brand Impact:** Car brand is one of the strongest determinants of price, with luxury brands commanding premium prices even for older models.
- **Age Depreciation:** Cars typically lose 15-20% of their value in the first year, with depreciation slowing down after 5 years.
- **Kilometer Effect:** Each additional 10,000 kilometers driven reduces a car's value by approximately 5-8%, with the effect diminishing for cars that have already crossed 100,000 kilometers.
- **Fuel Type Variations:** Diesel cars are generally priced 10-15% higher than equivalent petrol models, but this gap is narrowing in recent years.
- **Transmission Preference:** Automatic cars command a 15-20% premium over manual cars of the same model and age.
- **Geographic Variations:** Metropolitan cities have higher average prices, but also offer more variety in terms of available brands and models.
- **Value Leaders:** Certain brands like Maruti Suzuki, Hyundai, and Toyota offer the best combination of reliability and value retention in the Indian market.

7.2 Future Work

While this project provided comprehensive insights into the used car market in India, there are several opportunities to enhance the analysis in the future:

- **Expand the Dataset:** Include more details about car specifications, service history, and accident records to provide deeper insights into factors affecting price.

- **Sentiment Analysis:** Incorporate user reviews and ratings to understand perception about different car models and brands.
- **Predictive Modeling:** Develop machine learning models to predict future car values and optimal selling times.
- **Market Trend Analysis:** Track price changes over time to identify seasonal variations and long-term trends in the used car market.

7.3 Summary

This project has effectively mapped the used car market in India, offering detailed insights into price variations based on various factors like brand, age, kilometers driven, and fuel type. By adopting a data-driven approach, the study provides valuable information for car buyers, sellers, and market analysts, enabling them to make informed decisions.

The analysis revealed that while brand and model are primary price determinants, factors like kilometer reading, fuel type, transmission, and location significantly influence a car's market value. The interactive dashboards created as part of this project make it easier for users to navigate this complex market and find options that best suit their requirements and budget.

Through this comprehensive analysis, potential buyers can now identify cars that offer the best value for money, while sellers can better understand how to price their vehicles competitively in the market. The project demonstrates the power of data analytics in bringing transparency and informed decision-making to the used car market in India.